SDR U-NII-4 Interference Estimator and Experiments in a Real-World V2X Deployment

William Walker
Software Engineering
Kennesaw State University
Marrietta, GA, USA.

Sam Amoah
Software Engineering
Kennesaw State University
Marrietta, GA, USA.

Grayson Hatcher
Electrical and Computer Engineering
Kennesaw State University
Marrietta, GA, USA.

Khason Murphy
Software Engineering
Kennesaw State University
Marrietta, GA, USA.

Billy Kihei, Ph.D.
Electrical and Computer Engineering
Kennesaw State University
Marrietta, GA, USA.
https://orcid.org/0000-0002-0426-3145

I. Abstract

Unlicensed National Information Infrastructure (U-NII) bands, particularly U-NII-4, are forms of wireless communication that are growing in use. With more usage of U-NII-4, interference in this band can cause a problem with network performance. This paper presents a MATLAB Graphical User Interface (GUI) created to assist diagnosing interference issues by detecting and locating U-NII-4 interference. This system uses signal processing, spectrum analysis, GPS user tracking, data logging, and an antenna control system that allows users to calculate an estimate amount of interferes and log location of U-NII-4 interference.

II. Introduction

With increasing use of Unlicensed National Information Infrastructure (U-NII) devices operating within the 5.9 GHz band has raised natural concerns about the devices experiencing interference. The Federal Communications Commission (FCC) has recently issued a Notice of Proposed Rulemaking (NPRM) that proposes two interference mitigation approaches: Re-channelization and Detect and Vacate. Other research has been done on the mitigation technique of re-channelization [1]. However, this paper aims to show detecting and locating interference is an easy and possible solution.

Detect and Vacate is generally seen as the better approach because specification does not need to be changed on U-NII and Dedicated Short Range Communication (DSRC) devices. Detecting this interference is seen as an important aspect of maintaining secure communications between the devices. As shown in research conducted by the University of British Columbia [2] one of the main ways of allowing devices to read signal through interference is to change the detection threshold. We have focused on finding a way detect interference which permits personnel in the field to locate the sources and shut them down.

III. Existing Literature

A paper from Virginia Tech [3] verifies the impact that Wi-Fi bands operating in adjacent bands can have on the performance and reliability of C-V2X operations. The paper details various means of co-channel coexistence, including the use of detect and vacate in certain scenarios. This protocol would require Wi-Fi devices to use carrier sensing to detect DSRC transmissions, before connecting for the first time. They detail the requirements to use the channel, including a maximum duration of packets as well as a set AIFS of 300 used if the channel is unoccupied. In the event that a DSRC frame is detected during operation, the device must clear all ITS channels for ten seconds. In order to reconnect, the device will need to reconnect using the same procedure.

Along with evidence from our previous studies [4] showing that DSRC devices are more susceptible to interference from U-NII-4 devices passing by Roadside Units (RSUs). With results show RSUs losing up to 80% of Packet Reception Rate (PRR) from passing interference.

Plenty of research has been done investigating ways to allow DSRC devices to detect and report interference, as seen again by two papers published by the University of British Columbia [5] that gives a practical method of configuring DSRC networks to detect and report interference using capabilities already incorporated into the IEEE 802.11p standard or through a dedicated receiver.

Alongside [6] a proposed scheme that gives a modest cost to incorporate an interference detection and reporting system. Which is done by implementing a congestion and interference detection algorithm that will clone a second instance of the subsystem of the Security Credential Management System and use that to deliver reports to a spectrum misbehavior authority.

Furthermore, understanding how multiple devices compete for channel resources in unlicensed bands is important when addressing interference challenges. For example, A study on ML-based estimation [7] of the number of devices in industrial networks using unlicensed bands presents a unique approach to estimating device activity.

Another study on error-based interference detection in Wi-Fi networks [8] demonstrates how inter-technology interference can be recognized using commodity Wi-Fi devices by monitoring receiver error statistics.

Additionally, a research study on interference detection and resource allocation in LTE Unlicensed (LTE-U) systems with carrier aggregation [9] adopts logistic regression to train a classifier model for identifying users susceptible to interference from Wi-Fi.

Moreover, using a semantic cognitive enhancement network (SCEN) [10] for RFI signature detection introduces a method to detect RFI artifacts. SCEN employs an encoder-decoder architecture that integrates Atrous Spatial Pyramid pooling, depth-wise convolution, and a self-attentional mechanism, enabling pixel-wise RFI recognition Our research focuses more on giving those in the field searching for the interference an easy way to locate where the interference is coming from.

IV. Methods

The goal of our study was to test a number of different RF technologies for how they could affect PRR of an RSU when applied to them. Our driving hypothesis is that the stock RSU antennas paired with the powerful low noise amplifiers (LNA) of the RSUs, are raising the noise floor which allows U-NII-4 interference to affect the V2X PRR operation.

A. Real-Time Detector and Estimator for U-NII-4

The purpose of the estimator is to use an algorithm to analyze the raw signal data coming in through an SDR (software defined radio). Measuring the unii4 band while focusing on the amount of power seeping through the spectrum mask into the DSRC (Dedicated Short-Range Communications) band of V2X (vehicle to everything). It is important that we can measure this data to understand how it affects the reliability of V2X communications.

1) Detector and Estimator Formulation: The initial approach involved using Adjacent Channel Power Ratio (ACPR) as a method to estimate the amount of power bleeding into the main channel [11]. This was calculated by dividing the RF power in the adjacent channel (U-NII-4 Ch. 177) by the RF power in the main channel (DSRC Ch. 180). The resulting ACPR values were then organized into a matrix of varying sizes to compute a moving average. However, this method had fundamental flaws, as it was unable to effectively handle the rapid and dynamic changes present in the incoming data stream from the SDR. While this approach could detect interference, it failed to provide an accurate representation of the interference levels, as shown in Figure 1.

Although some changes in ACPR could be observed, the data exhibited spikes in the opposite direction of expected trends. The high level of noise in the ACPR data made it difficult to programmatically quantify the interference. Due to these limitations, the focus shifted from measuring the amount of interference to detecting individual instances of U-NII-4

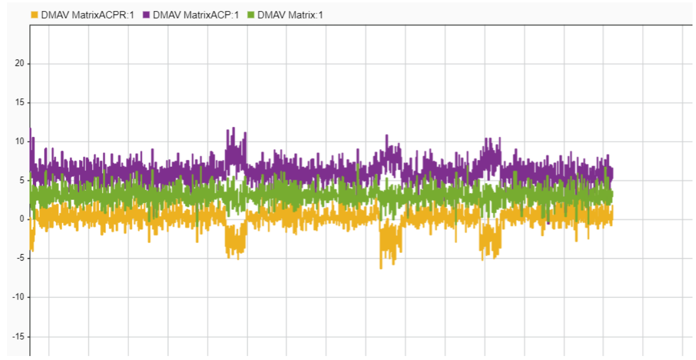

Fig. 1. ACPR data as an interference estimation is too noisy.

packets, using a time scope to ensure that each "sample batch" of interference could be identified.

Building on this knowledge, the process transitioned to working directly with raw data at a real-time rate. The magnitude of the complex signals was computed first, isolating signal power from the raw data and making it easier to quantify and analyze interference levels [12]. This output was passed through a windowed integrator, summing the signal over a specified window to smooth the power fluctuations and provide an averaged view of the interference levels. The next step involved taking the absolute value of the signal to prevent negative values, followed by squaring the signal to emphasize higher power levels. This signal processing chain served as the foundation for the final flowchart, see Figure 2

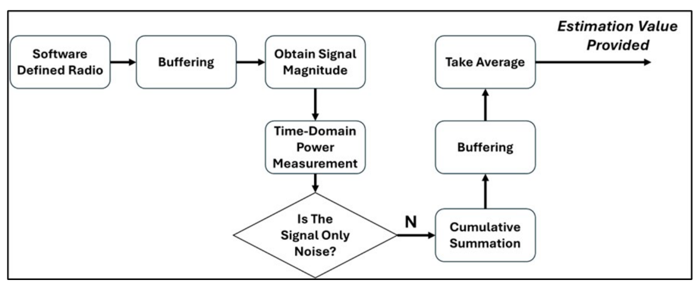

Fig. 2. Final flowchart for detecting and estimating U-NII-4 interference activity in Ch. 177.

To begin this process, the estimator collects .04 seconds worth of data which is 1 million samples in a buffer to allow for processing. After collecting the data which consist of real and imaginary numbers, the real and imaginary numbers will then convert into a complex number by finding the magnitude squared of the set of 1 million samples the data converts, using the equation:

$\sum |U[n]|^2$

From there the data is processed to distinguish interference from noise by applying a thresholding. The threshold was determined through lab testing by analyzing the maximum values of the magnitude of u. These values were plotted on a time scope, allowing for the identification of a consistent threshold as shown in Figure 3. The threshold represented as (t) = 2.5 * 103 points below which U-NII-4 interference

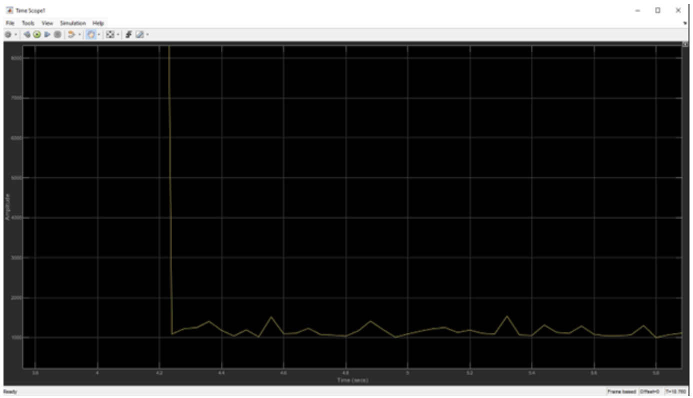

Fig. 3. Determining the noise threshold for the estimator flowchart for the SDR used in this study.

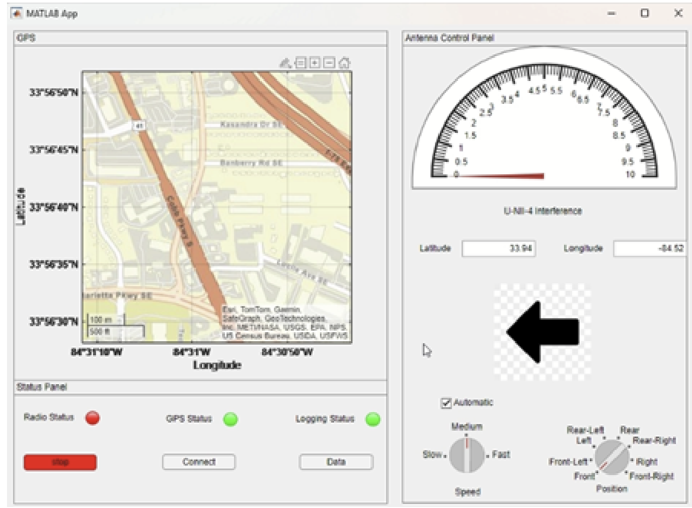

Fig. 4. GUI main user control screen

signals do not drop, and above which noise signals do not rise, providing a clear boundary. This threshold is then integrated into the estimator, dynamically filtering the data. Signal values below the threshold are classified as noise and set to zero, while values above the threshold are retained as interference. For values above the threshold, normalization is applied using the equation:

$$y = \frac{y}{\max(u)}$$

This provides context to the results by setting the maximum possible value to 1. Following this, the cumulative sum of the normalized values is computed, providing an end value that is contextualized on a scale of 0 to 1 million, corresponding to the frame size of 1 million samples. To obtain a better understanding of the interference over time, 12 end points are buffered and averaged. This yields the average power over the last half second, smoothing the data and ensuring that the results accurately reflect dynamic levels of interference.

2) GUI: On start up MATLAB creates the instance of the GUI and runs a start-up function. Within this function global variables for the system are initialized and set to default values, this is also when the MATLAB GUI, see Figure 4 will connect with the Simulink attached to the GUI. When the GUI instance is created buttons for connecting the GPS receiver, and Antenna control system are made available. After the connection button is activated, MATLAB searches all active serial ports and connects to the GPS receiver first then connects to the Arduino UNO on the Antenna apparatus.

The user must then set up a file for data logging. By using a separate screen, a user must type in a name for the file, then the system will automatically create a text file with the name given.

After the peripherals have been connected the start button becomes available, when activated the entire system will begin operating. Within the start function the main function of the system is set up and begins to run.

3) GUI Main Function: The main function of the GUI is a timer object which is constantly being set off in one second intervals. When the object is activated, it runs an update function for the GUI. This Update function is set to run every time the timer object activates, and within has a list of smaller functions that will run the system.

4) GPS Receiver: First in the list is getting latitude and longitude from the GPS receiver and plotting it on the map in the GUI. This is done by reading NMEA (National Marine Electronics Association) data, each NMEA string has values for a time stamp, latitude, longitude, a quality indicator, and many more. By simply parsing the string the system can extract latitude and longitude. Then by using a MATLAB add on that comes with a map function, the system will plot the map to the newly received coordinates.

5) Data Logging: Next the system will check if the user has decided to log system data automatically. If the user has then it will run a data log function that will encapsulate the data in a CVS format. There are five variables within the CVS string data: Latitude, longitude, U-NII-4 interference, direction from user, and a time stamp.

6) Antenna Controls: The system will allow users to rotate an antenna in 360 degrees, split into 8 directions around the user:

- Front: position 0, 0 degrees (origin)
- Front-Left: position 1, 45 degrees
- Left: position 2, 90 degrees
- Rear-Left: position 3, 135 degrees
- Rear: position 4, 180 degrees
- Rear-Right: position 5, 225 degrees
- Right: position 6, 270 degrees
- Front-Right: position 7, 315 degrees

If the user decides to rotate the antenna automatically there are two setting for the speed: fast, and slow.

B. Antenna Apparatus

1) Arduino Functions: Once the on-board Arduino is connected to the MATLAB program, a start-up function is activated which sets up the pins to control the motor holding the antenna. After this an origin function starts that will set the antenna to position 0 (facing forward). The motor will spin slowly until a sensor is triggered at the front of the antenna apparatus; this process is done to account for motor drift.

When the GUI sends a new position to the antenna control system, it takes the current position and the target position and multiplies them by 50(the number of motor steps in between each position). The system will then perform calculations to determine how many steps are needed to achieve the desired position.

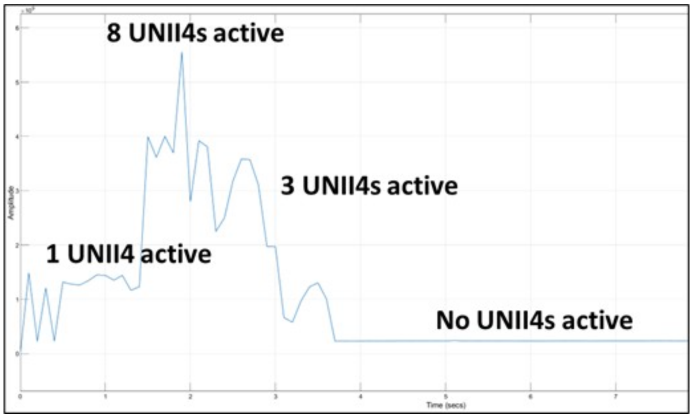

Fig. 5. The estimator is able to detect and estimate the number of U-NII-4 devices in a lab setting.

2) 3D Modeling of Apparatus:

V. Results

A. In Lab Validation of Detector and Estimator

When in a lab setting, the estimator/detector was very stable, and could show the number of estimators during time of the experiments, see Figure 5. The lab results shown in Figure 5 demonstrate the effectiveness of the final approach of the estimator in detecting and quantifying interference in the U-NII-4 band. The lab data shown highlights the magnitude of interference, providing valuable insights into its behavior over time. Overall, the results confirm that the estimator is effective in detecting and quantifying interference in the U-NII-4 band. By leveraging real-time processing, normalization, and threshold-based detection, the system provides a robust framework for analyzing interference and its impact on CV communication.

B. War Drive of a Real-World V2X Deployment

The estimator was taken on a war drive near select midtown and downtown RSU locations (both planned and installed), supplied by GPS locations from GDOT. There was no significant interference, however, near the Georgia World Congress Center area, a burst of unidentified high-power interference was observed (could be communications or radar systems). It is intermittent so it is with low likelihood that this interference disrupts CV applications.

VI. Conclusion

High performance software defined radio technology can be used with an antenna assembly and MATLAB SIMULINK to reliably detect Wi-Fi 5 interference (among other interference). To implement these findings: an interference detection kit is provided complete with hardware and software that GDOT can use to locate interference, without needing to hire a third-party firm to locate and identify interference. The current software version is not optimized and often requires a restart and is a solid foundation to build from now that the signal processing chain is discovered.

VII. ACKNOWLEDGMENTS

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